

Policy incentives, market competition, and firms' international strategy choices: Evidence from traditional Chinese medicine enterprises

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Abstract: With the strong support of the "Traditional Chinese Medicine Going Global" national strategy, how to actively promote the internationalization of Traditional Chinese Medicine (TCM) enterprises is an important issue that urgently needs to be explored. Since the internationalization motives and capabilities of Traditional Chinese Medicine (TCM) enterprises are different from those of Developed Market Multinational Enterprises (DMNEs) and Emerging Market Multinational Enterprises (EMNEs), existing research on the explanatory role of policy support and industry competitive pressure cannot fully interpret the international strategic choices of these enterprises. To explore this new research issue, this paper discusses the differentiated impact mechanisms of policy support and industry competition intensity on the international expansion of TCM enterprises based on the push-pull theory, and further analyses the moderating roles of marketing capabilities and innovation capabilities of the enterprises. This paper analyses all A-share listed enterprises in China's Traditional Chinese Medicine (TCM) industry from 2012 to 2021 as a research sample. The results show that both policy support and industry competition intensity positively affect the breadth of internationalization of TCM enterprises. Policy support has a stronger driving effect on the internationalization of TCM enterprises with weaker capabilities, while industry competition intensity has a stronger driving effect on the internationalization of TCM enterprises with stronger capabilities. This paper makes a significant contribution to the existing research on the motivations for internationalization and provides a new theoretical interpretation for the strategic choices of TCM enterprises in the

context of cultural differences within TCM. Furthermore, it provides theoretical guidance and empirical evidence for the practice of TCM enterprises going overseas and the formulation of industry policies.

Keywords:

TCM cultural differences; policy support; industry competition intensity; enterprise internationalization

1 Introduction

The internationalization of traditional Chinese medicine (TCM) enterprises is one of the key measures for expanding the global influence of TCM culture and enhancing China's discourse power and national soft power in global health governance, and therefore has received strong support at the national strategic level. However, over the past two decades, the internationalization of TCM enterprises has lagged significantly behind that of other industries. According to Lee's (1966) push—pull theory, the internationalization motives of emerging market multinational enterprises (EMNEs) can be summarized as a combined consideration of push factors (such as home-country institutional deficiencies and reform and opening-up) (Buckley et al., 2007; Jormanainen & Koveshnikov, 2012), pull factors (such as host-country technology and market development) (Dunning & Lundan, 2008), and intervening obstacles (such as cultural distance and institutional distance). Kalotay and Sulstarova (2010) argue that the international expansion of EMNEs is generally driven by favorable institutional environments in host countries. In contrast, the internationalization motives of TCM enterprises

exhibit two characteristics that differ markedly from the conclusions of mainstream research. Firstly, TCM enterprises face severe cultural differences in the process of internationalization, which are manifested in formal institutional differences (the varying degrees of international recognition of the legal status of TCM) and informal institutional differences (the lack of widespread acceptance of its therapeutic value in international markets) . Secondly, unlike EMNEs at the developmental stages described in the linkage—leverage—learning (LLL) theory and the springboard perspective, TCM enterprises do not choose international expansion in order to seek strategic assets; instead, they are more inclined to apply their unique medical knowledge and technologies to the development and expansion of host-country (or regional) markets. Given that the internationalization motives of TCM enterprises deviate from traditional international business theoretical paradigms, it is necessary to seek new explanations to reveal their distinctive strategic choices that differ from those of other EMNEs.

Based on push—pull theory, in the context of the internationalization of traditional Chinese medicine (TCM) enterprises, on the one hand, cultural differences related to TCM constitute a critical "intervening obstacle." This obstacle makes it difficult for TCM enterprises to obtain market-related "pull" forces from host countries. On the other hand, although TCM enterprises possess distinctive medical knowledge and technological advantages, these same characteristics also weaken the technology-related

"pull" forces originating from host countries. Consequently, the internationalization of TCM enterprises relies heavily on "push" forces from the home country to overcome intervening obstacles and to tap into the potential pull forces of host countries. Among push factors, policy support and industry competitive intensity both play significant roles (Zhang & Zhu, 2021). On the one hand, in 2016 the State Council promulgated the "Outline of the Strategic Plan for the Development of Traditional Chinese Medicine (2016—2030)", elevating the promotion of the overseas development of TCM to the level of a national strategy. In addition, the "Development Plan for Traditional Chinese Medicine under the Belt and Road Initiative (2016—2020)", jointly formulated by the National Administration of Traditional Chinese Medicine and the National Development and Reform Commission, has provided strong policy guarantees for enterprises to further explore markets along the Belt and Road. On the other hand, as industry competition intensifies, profit margins in the domestic market have gradually narrowed (Zhang & Zhu, 2021), forcing TCM enterprises to regard internationalization as a means of strategic adjustment in order to seek new growth opportunities and market space (Leonidou et al., 2022) and to ensure sustained profitability. However, how policy support and competitive pressure generate distinct pushing mechanisms in the international expansion of TCM enterprises has not yet been clearly theorized or rigorously examined empirically. Against this backdrop, an in-depth

investigation of the differentiated pushing mechanisms of policy support and competitive pressure, as well as how TCM enterprises leverage their own capabilities to make internationalization strategy choices, is of significant theoretical and practical importance.

Drawing on push—pull theory, this study incorporates both push factors and firm-specific capabilities that influence the internationalization of traditional Chinese medicine (TCM) enterprises into an integrated analytical framework of internationalization motives. It aims to examine, under conditions of TCM-related cultural differences, the differentiated pushing mechanisms generated by policy support and industry competitive intensity, as well as how TCM enterprises with varying levels of innovation capability and marketing capability make different internationalization strategy choices. Firm internationalization generally encompasses two distinct dimensions: depth and breadth, which differ substantially from each other. Internationalization depth refers to the proportion of overseas sales to total sales or the proportion of overseas assets to total assets, reflecting a firm's degree of dependence on foreign markets. In contrast, internationalization breadth refers to the number of countries or regions a firm has entered (Schwens et al., 2018), capturing the diversity and complexity of national differences that the firm faces. Compared with an increase in internationalization depth, an expansion in internationalization breadth implies that firms are exposed to more complex and heterogeneous national

and regional environments, which in turn intensifies TCM-related cultural differences and potentially amplifies the role of home-country "push" factors. Accordingly, this study adopts internationalization breadth as the outcome variable and empirically tests the proposed hypotheses using panel data from publicly listed TCM enterprises over the period 2012–2021.

The marginal contributions of this study are threefold. Firstly, owing to TCM-related cultural differences, the internationalization motives of traditional Chinese medicine (TCM) enterprises differ substantially from those of previously studied EMNEs, and existing research has been unable to provide an adequate explanation for this divergence. Building on push–pull theory and focusing on the unique context of TCM enterprise internationalization, this study proposes an analytical framework for the mechanisms underlying TCM enterprises' internationalization motives. In particular, it highlights the differentiated pushing mechanisms of home-country policy support and industry competitive pressure, thereby offering new insights and extending the existing literature. Policy support helps enhance the recognition and acceptance of TCM in international markets, thereby alleviating the "intervening obstacles" caused by TCM-related cultural differences and strengthening the market-related "pull" forces from host countries. By contrast, industry competitive intensity does not directly alter host-country "pull" forces or the intervening obstacles arising from TCM-related cultural differences; rather, it compels TCM

enterprises, through market mechanisms, to proactively explore international market opportunities (Buckley et al., 2023). Secondly, this study finds that TCM enterprises with different capability endowments respond to policy support and industry competitive pressure with differentiated strategic choices in their international expansion. Enterprises with stronger innovation and marketing capabilities are better able to overcome TCM-related cultural differences, thereby achieving the cross-border transfer and exploitation of competitive advantages. In contrast, firms with relatively weaker capabilities, which are unable to rely on their own resources to effectively overcome the intervening obstacles posed by TCM-related cultural differences, tend to exhibit greater dependence on policy-driven forces. Overall, this study provides empirical evidence to inform both the internationalization strategy choices of TCM enterprises and the formulation of industry policies.

2 Theoretical background and hypothesis development

2.1 Characteristics of the internationalization of traditional Chinese medicine (TCM) Enterprises

Compared with multinational enterprises from developed countries (DMNEs) and emerging market multinational enterprises (EMNEs), the internationalization of traditional Chinese medicine (TCM) enterprises exhibits distinctive characteristics. Firstly, with respect to cultural differences, although TCM enterprises, like DMNEs and EMNEs, face

cross-national cultural differences (Brouthers & Brouthers, 2001; Tihanyi et al., 2005), their unique medical system exposes them to more pronounced cultural disparities. Moreover, they must also confront differences in industry regulations, thereby creating a novel research context (Figure 1). He et al. (2023) analyzed cultural differences in TCM and argued that they are jointly constituted by informal institutional differences (national cultural differences) and formal institutional differences (industry regulatory differences). National culture profoundly influences individuals' social values, behavioral patterns, and cognitive frameworks (Slangen & van Tulder, 2009). Cultural differences between countries lead to substantial divergences between Western medicine and TCM in terms of cultural cognition, medical philosophy, theoretical foundations, and therapeutic approaches. At the same time, regarding differences in industry regulations, the legal status of TCM has not yet been widely recognized in the international community, and substantial variation exists across countries and regions in the extent to which TCM is regulated and accepted. Taking the United States as an example, although acupuncture has been granted legislative recognition in 47 states and one federal district, the legality of other TCM therapeutic practices and the use of herbal medicines has not been established within a unified legal framework. Consequently, the internationalization process of TCM enterprises has progressed more slowly than that of Chinese enterprises in other industries.

Secondly, traditional Chinese medicine enterprises also exhibit distinct characteristics in terms of capabilities. Existing research indicates that Domestic Market-Driven New Enterprises (DMNEs) typically possess strong technological and innovative capabilities, adopting a market-seeking orientation and leveraging ownership advantages for overseas expansion (Dunning, 1995). Enterprise Market-Driven New Enterprises (EMNEs), generally lacking ownership advantages, tend to pursue a strategic asset-seeking orientation. They often possess extensive marketing experience and typically employ a 'stepping stone strategy' for international expansion. Similar to DMNEs, TCM enterprises pursue market-seeking objectives. During internationalization, they prioritise integrating their core strengths—unique medical knowledge systems, traditional diagnostic and therapeutic techniques, and extensive clinical experience—with host country (or region) market demands to expand internationally. Mirroring EMNEs, TCM enterprises also possess outstanding marketing capabilities.

From the perspective of the push-pull theory, the drivers of corporate internationalization can be attributed to the combined effects of push factors (home country policy support and industry competitive pressures), intermediate barriers (cultural and institutional differences), and pull factors (the potential for technological advancement and market development in host countries). Given the unique characteristics of traditional Chinese medicine (TCM) enterprises' internationalization, push factors provide

crucial support for their global expansion (Utesch-Xiong et al., 2024). while intermediary barriers are particularly pronounced due to the distinctiveness of TCM cultural differences. These not only impact market acceptance in host countries but also heighten uncertainty and risk within the internationalization process, rendering the already insufficient "pull" from host countries even more challenging to access.

[Insert Figure1 about here]

2.2 Policy support and the breadth of internationalization

Policies reflect a country's strategic intentions and bear the critical responsibility of allocating resources and achieving strategic objectives. Policy support plays a vital role in the process of firm internationalization: it not only provides favorable external conditions for firms but also exerts a key influence on shaping their internationalization strategies and behaviors (Reimann et al., 2018; Gammeltoft & Panibratov, 2024). Some scholars argue that government policies constitute an important driving force guiding the internationalization behavior of emerging market multinational enterprises (EMNEs), significantly accelerating their international expansion (Utesch-Xiong et al., 2024) and exerting a positive impact on their internationalization outcomes (Cui et al., 2022). In advancing their internationalization strategic objectives, traditional Chinese medicine (TCM) enterprises have likewise received policy support and guidance.

Firstly, government support can, to a certain extent, mitigate the

"intermediate obstacles" faced by TCM enterprises in their internationalization process. On the one hand, policy support provides clear guidance for the internationalization of TCM enterprises and helps establish a favorable policy environment, thereby reducing the costs imposed by institutional differences (Cui et al., 2022). For example, the Outline of the Strategic Plan for the Development of Traditional Chinese Medicine (2016–2030) issued by the State Council in 2016 explicitly proposed key tasks such as "implementing an overseas development initiative for TCM and promoting the global expansion of TCM technologies, medicines, standards, and services." In addition, measures such as streamlining export approval procedures for TCM products have reduced administrative complexity and improved the efficiency of outward investment by TCM enterprises. On the other hand, policy support facilitates the international dissemination of TCM culture, thereby enhancing host-country markets' understanding of TCM. As overall awareness and evaluation of TCM culture in host-country markets improve, acceptance of TCM products correspondingly increases, which helps to reduce cultural distance. Therefore, policy support weakens the "intermediate obstacles" encountered by TCM enterprises in the course of internationalization.

Secondly, policy support helps cultivate potential "pull forces" in host-country markets, thereby stimulating the willingness of TCM enterprises to "go global." On the one hand, the dissemination of TCM

culture enhances host-country markets' awareness and acceptance of TCM products, generating latent demand for TCM-related goods such as medicinal materials, patent medicines, and health products. Such demand constitutes a form of "pull force" that attracts TCM enterprises to enter host-country markets in order to meet local needs. On the other hand, from a firm self-selection perspective, enterprises that receive policy support possess stronger "motives" and "capabilities" to implement policy initiatives. Policy instruments such as tax incentives and government subsidies can enhance the competitiveness of TCM enterprises, thereby indirectly granting firms greater autonomy in strategic choices. This form of policy support also enables firms to identify and exploit potential demand in host-country markets more effectively.

In summary, policy support not only mitigates the "intermediate obstacles" arising from cultural and institutional differences, but also contributes to the cultivation of host-country markets and the formation of potential "pull forces" for TCM enterprises, thereby promoting their outward internationalization activities. Based on the above analysis, this study proposes:

Hypothesis1: Policy support promotes the breadth of internationalization of
TCM enterprises.

2.3 Industry competitive intensity and the breadth of internationalization

Competitive intensity refers to the extent to which firms within the same industry exert competitive pressure on one another by attacking rivals' competitive positions in order to enhance their own performance (Giachetti, 2023). In international business research, competitive pressure in the home market has been shown to influence firms' internationalization decisions—such as the level and pace of resource commitment—thereby exerting a significant impact on internationalization strategies (Wójcik et al., 2025; Fu et al., 2024). Moreover, intensified industry competition is often associated with increasing resource constraints faced by firms, as cross-border expansion consumes scarce organizational resources and the outcomes of internationalization strategies are contingent upon external environmental factors, including competitive intensity (Fu et al., 2024). In addition, when product differentiation declines and homogeneity increases, firms are more likely to engage in intense rivalry centered on the pursuit of market share (Porter, 1979). From a market-based perspective, industry competitive pressure constitutes a key driver of firms' outward foreign direct investment. In highly competitive market environments, firms often regard internationalization as a strategic means of overcoming domestic growth constraints and expanding market share (Wu et al., 2016; Tang & Buckley, 2022). Buckley et al. (2023) further argue that intensifying competition in domestic industries compels firms to seek new development opportunities abroad. From a cost-based perspective, heightened industry competition

directly erodes firms' profit margins in the domestic market, leading to declining profitability and thereby forcing firms to pursue new sources of profit growth in international markets (Leonidou et al., 2022). This compression of domestic profit space, in turn, accelerates firms' international expansion.

For traditional Chinese medicine (TCM) firms, the industry in China is characterized by a large number of firms, low market concentration, and intense competition. Under such conditions of high competitive intensity, TCM firms must seek new opportunities and markets through internationalization in order to survive and sustain growth. Unlike policy support, which functions as an enabling or facilitating force, industry competitive intensity acts as a push factor in firms' internationalization processes: it does not directly shape the pull factors of host-country markets, nor does it alleviate the intermediate barriers encountered during international expansion. Intensifying competition in the domestic market compresses profit margins and heightens competition for market share among TCM firms. To maintain profitability and growth, firms are therefore compelled to search for new market opportunities beyond national borders. Under such competitive pressure, TCM firms are driven to explore and further develop international markets. Consequently, industry competitive intensity constitutes a direct push force for the internationalization of TCM firms. Accordingly, a positive relationship is expected between industry

competitive intensity and the breadth of internationalization of TCM firms: the higher the level of competitive intensity, the greater the breadth of firms' internationalization. Based on this reasoning, this study proposes the following hypothesis:

Hypothesis2: Industry competitive intensity positively influences the breadth of internationalization of traditional Chinese medicine firms.

2.4 The Moderating Role of Firm Capabilities

According to dynamic capability theory, firms must dynamically adjust their strategies based on changes in the external environment and their own firm-specific capabilities in order to sustain competitive advantage. Similar to developed-market multinational enterprises (DMNEs), traditional Chinese medicine (TCM) firms possess distinctive innovation capabilities; at the same time, akin to emerging-market multinational enterprises (EMNEs), they exhibit strong marketing capabilities. For example, to enhance organizational legitimacy and address formal institutional differences, Tasly leveraged its innovation capabilities to obtain approval from the U.S. Food and Drug Administration (FDA), thereby seeking entry into global markets through one of the most stringent pharmaceutical regulatory systems. To narrow value gaps arising from informal institutional differences, Tongrentang adopted a "culture-first" marketing strategy, actively exploring new approaches to serving overseas markets and gaining cultural legitimacy and value recognition. As discussed above, policy support and industry

competitive pressure promote the internationalization of TCM firms through distinct mechanisms. However, firms' responses to these push forces may vary depending on their marketing and innovation capabilities. Accordingly, this study examines the moderating effects of these capabilities on TCM firms' internationalization strategy choices under the influence of policy support and industry competitive pressure.

2.4.1 The Moderating Effects of Marketing Capability and Innovation Capability on the Relationship between Policy Support and Internationalization Breadth

Marketing capability is widely regarded as a critical capability enabling emerging-market multinational enterprises to establish and expand relational networks in international markets. It helps firms identify international opportunities, reduce perceived uncertainty, and better understand new market contexts (Hunt & Madhavaram, 2020). Chinese firms generally demonstrate strong capabilities in leveraging distribution channel resources and maintaining relational networks. Such guanxi-based organizational networking capabilities are widely employed in the Chinese business context and are shown to be significantly associated with firms' market expansion, competitive positioning, and enhanced organizational performance (Park & Luo, 2001; Luo et al., 2012). Within distribution channel contexts, guanxi is also regarded as a critical resource for establishing and sustaining buyer—seller relationships, and under certain conditions, it has the potential to

serve as a source of sustained competitive advantage (Wiegel & Bamford, 2015). In complex international environments, where “intermediate barriers” arise from cultural differences associated with traditional Chinese medicine, TCM firms with weaker marketing capabilities are often unable to effectively address challenges related to legitimacy and trust through their own capabilities. Consequently, such firms rely more heavily on policy support and guidance to mitigate intermediate barriers and cultivate potential overseas markets in order to expand internationally. In contrast, firms with strong marketing capabilities possess clear marketing advantages. When confronted with intermediate barriers such as market entry restrictions and cultural differences, these firms are more likely to rely on their own marketing strengths to overcome challenges rather than depend on policy support (Yamin & Harhoff, 2010). Therefore, marketing capability can substitute for the role of policy support in promoting internationalization breadth. In other words, when TCM firms possess strong marketing capabilities, the “push” effect of policy support on internationalization is weakened.

Innovation capability is one of firms’ core competitive advantages, enabling them to proactively adapt to and shape changes in competitive environments (Vrontis & Christofi, 2021). Compared with TCM firms with strong innovation capabilities, firms with weaker innovation capabilities often suffer from deficiencies in technological development and market

expansion experience (Kastelli et al., 2024). Their relatively weak competitiveness makes it difficult for them to expand internationally based on internal capabilities alone. Such firms therefore tend to rely on policy support to overcome intermediate barriers and explore new markets, gradually accumulating internationalization experience. For TCM firms with weaker innovation capabilities, policy support thus plays a particularly important role in shaping internationalization strategies. By contrast, innovative traditional Chinese medicine (TCM) enterprises typically possess both strong technological capabilities and robust market development competencies, enabling them to more effectively address legitimacy challenges arising from differences in overseas regulatory regimes, quality standards, and evidentiary requirements. As a result, these firms can enhance the efficiency of their internationalization processes and increase the likelihood of successful market entry (Morgan et al., 2012; Lin et al., 2018; Li et al., 2018; You et al., 2022). Firms with strong innovation capabilities are more likely to rely on their own technological advantages and market competitiveness to proactively address intermediate barriers and cultivate overseas markets, rather than depend on policy support. For instance, Tongrentang's successful presence in international markets is largely attributable to its technological innovation and stringent product quality control, which have enabled its products to meet pharmaceutical manufacturing standards in multiple countries and regions, including FDA

certification in the United States. In essence, firms with strong innovation capabilities exhibit a reduced dependence on policy support, thereby substituting for its role in promoting internationalization. Accordingly, innovation capability weakens the positive effect of policy support on internationalization breadth. Based on the above analysis, this study proposes the following hypothesis:

Hypothesis3: Compared with firms possessing strong innovation and marketing capabilities, policy support exerts a stronger positive effect on the breadth of internationalization for traditional Chinese medicine firms with weaker innovation and marketing capabilities.

2.4.2 The moderating effects of marketing capability and innovation capability on the relationship between industry competitive intensity and the breadth of internationalization.

Industry competitive pressure not only compels firms to pursue innovation and efficiency improvements in their domestic markets but also incentivizes them to seek growth opportunities abroad in order to diversify risks and enhance returns (Hu & Lyu, 2024). However, as Porter (1979) emphasizes, heightened competitive intensity places substantial demands on firms' resources and capabilities during the process of international expansion. Importantly, industry competitive pressure alone cannot mitigate the liability of foreignness faced by traditional Chinese medicine (TCM) firms, nor can it facilitate the development of foreign market demand. Consequently, firms'

internationalization outcomes depend critically on their internal capability endowments. Marketing capability represents a key firm-level resource that conditions the extent to which competitive pressure can be translated into successful international expansion. Firms with strong marketing capabilities are better able to articulate and communicate the core value of their brands and products, thereby reducing uncertainty and risk in foreign market entry. Prior research demonstrates that adaptive and dynamic marketing capabilities significantly enhance firms' international performance by enabling them to respond effectively to environmental turbulence and competitive intensity in overseas markets (Reimann et al., 2022; Pfajfar et al., 2024). Under intensified competitive pressure, firms possessing superior marketing capabilities are therefore more likely to leverage these advantages to proactively expand into international markets. In contrast, firms with weaker marketing capabilities often lack the resources and competencies necessary to cope with the complexity of international markets, limiting their ability to overcome the liability of foreignness and exploit new market opportunities. Accordingly, the positive effect of industry competitive pressure on the breadth of internationalization is contingent upon firms' marketing capability.

Under conditions of high industry competitive intensity, firms face heightened pressure to differentiate and seek new growth opportunities. Innovation capability enables firms to develop differentiated advantages

through superior R&D and technological adaptation, thereby reducing the risks of homogeneous competition and facilitating entry into new market domains. For traditional Chinese medicine (TCM) firms, strong innovation capability is particularly critical for overcoming cultural and institutional barriers associated with internationalization and for developing host-country markets. In contrast, firms with weaker innovation capability lack the technological resources necessary to respond effectively to intensified competition and to overcome the liability of foreignness. Accordingly, firms with stronger innovation capability are better positioned to translate domestic competitive pressure into broader international expansion.

Hypothesis 4: Industry competitive intensity has a stronger positive effect on the breadth of internationalization for TCM firms with stronger innovation and marketing capabilities than for those with weaker capabilities.

3 Methodology

3.1 Data Sources and Variable Selection

The sample consists of A-share listed firms in the traditional Chinese medicine (TCM) industry that engaged in internationalization activities during the period 2012–2021. Firm-level data were obtained from the China Stock Market and Accounting Research (CSMAR) database and firms' annual reports. The sample was identified based on the "Shenwan Industry Classification (2021 revised edition)", focusing on firms classified in the TCM

industry. The number of overseas subsidiaries was manually calculated using information disclosed in firms' annual reports and was cross-validated and supplemented with corporate news reports. Given that the policy variable is defined using 2017 as the cutoff year and that policy effects on internationalization breadth may exhibit a time lag, the internationalization breadth in the subsequent year was used as the dependent variable. Based on this consideration, the raw data were further screened and processed using the following criteria: (1) observations from firms designated as ST or *ST were excluded; (2) observations with missing values for key variables were removed; and (3) continuous variables were winsorized to mitigate the influence of extreme values. After applying these procedures, the final sample comprises 498 firm-year observations of listed TCM firms.

3.2 Variable definition and explanation

The dependent variable, internationalization breadth (Inbre), captures the scope and geographic dispersion of a firm's overseas market presence, reflecting its country-level diversity and operational complexity as a core dimension of international expansion. Extant research commonly employs a dual-dimensional framework to measure internationalization: depth (typically proxied by foreign sales/total sales or foreign assets/total assets) and breadth (generally measured by the number of countries/regions of operation) (Schwens et al., 2018). Compared to depth—which indicates market dependency—breadth more saliently reflects a firm's capacity to navigate the complexity of cross-cultural institutional environments. For Traditional Chinese Medicine (TCM) enterprises, expanding geographic scope amplifies the complexity of cross-national cultural distance and sectoral regulatory

divergence, thereby accentuating the efficacy of home-country institutional support in mitigating compounded TCM cultural barriers. Consequently, this study operationalizes internationalization breadth as the outcome variable, adopting the number of overseas subsidiaries, following Lu & Beamish(2004), to rigorously quantify this construct.

The first independent variable, Policy Support (Policy), is measured using a time-based dummy variable (He et al., 2022). Considering that the Development Plan for Promoting High-Quality Integration of Traditional Chinese Medicine into the Belt and Road Initiative (2016-2020) was promulgated in late December 2016, 2017 is designated as the first year of policy implementation. Accordingly, the policy variable is assigned a value of 1 for statistical years 2017 and onwards, and 0 otherwise.

The second independent variable, Industry Competition Intensity (Indlena), is operationalized using the industry Lerner index (Lerner,1934) . The industry Lerner index represents the degree of market monopoly power, where a higher value signifies weaker competition. Consequently, this inverse measure was reverse-coded before analysis to ensure higher values correspond to greater competition intensity.

The first moderator variable, Marketing Capabilities (MC), is measured using the year-end proportion of sales personnel in the total number of employees, following the approach established by Feng et al. (2015). ***The second moderator variable, Technological Capability (TC)***, is measured by R&D intensity (R&D expenditure divided by operating revenue), based on the methodology outlined by James et al. (2020).

We control for other variables potentially influencing firms' internationalization decisions. ***Firm size***, a significant organizational characteristic affecting strategic choices including internationalization (Hitt et al., 2006), is measured by the natural logarithm of total assets. ***Firm age***,

important for internationalization timing as it alters anticipated outcomes and risk levels in initial foreign investment (Sapienza et al., 2006), is measured as the natural logarithm of years from the statistical year to the firm's founding year + 1. Past performance, influencing resource bases and risk willingness (March & Shapira, 1987), is controlled using ***Return on Assets (ROA)***, ***Total Asset Turnover (TAT)***, and ***the Tangible Asset Ratio***. Financial slack (unabsorbed slack), affecting risk-taking behavior (Singh, 1986; Iyer & Miller, 2008), is measured by the ***Leverage Ratio (Total Liabilities/Total Assets)***. ***Ownership concentration***, impacting corporate strategy (Thomsen & Pedersen, 2000), is controlled using the percentage shareholding of the top ten largest shareholders. Furthermore, acknowledging that ownership structure influences international risk-taking, we additionally control for ***management age*** (average age of the management team), ***CEO/Chair duality*** (a dummy variable equal to 1 if the Chairman also serves as the CEO, 0 otherwise), ***board size*** (natural logarithm of the total number of board members), and the ***proportion of independent directors*** (ratio of independent directors to total board members).

4 Empirical results and analysis

4.1 Descriptive statistics and correlation analysis

Table 1 reports the descriptive statistics and correlation analysis of the variables. As shown in Table 1, the correlation coefficient between policy support and internationalization breadth is 0.102, and the correlation coefficient between industry competition intensity and internationalization breadth is 0.122, both below 0.6. Additionally, the variance inflation factors (VIF) for each model were examined. The VIF for policy support is 1.74 and

for industry competition intensity is 1.93, both of which are below 3. Therefore, it can be concluded that there is no severe multicollinearity issue in the model.

4.2 Regression Analysis Results

Before conducting regression analysis, the Hausman test was employed to determine the adoption of a fixed effects model. Table 2 reports the regression results. Model (1) represents the baseline model containing only control variables. Model (2) examines the direct effects of the policy on internationalization breadth and industry competition intensity on internationalization breadth. The results show that the regression coefficients for Policy and Indlena are 0.467 and 9.315, respectively, both statistically significant at the 1% and 5% levels with positive signs. In conclusion, Hypothesis 1 and Hypothesis 2 are supported.

Based on the previous analysis, both policy support and industry competition intensity significantly promote enterprises' internationalization. Given that firms' internationalization behaviors are inevitably jointly influenced by their home-country institutional environment and competitive intensity, this study incorporates the interaction term between policy support and industry competition intensity (Policy $\times$ Indlena) into the baseline regression model to examine their dual impact. The results are presented in Table 2. As shown in Column (7), the coefficient for the policy support and industry competition intensity interaction term is 17.02, which is statistically

significant at the 1% level with a positive sign. This indicates that the interaction effect between industry competition intensity and policy support has a significant positive impact on firms' internationalization breadth. This positive interaction effect suggests that policy support and industry competition intensity exhibit a complementary effect in promoting enterprise internationalization.

[Insert Table1 about here]

[Insert Table2 about here]

Table 2 presents the results for Models (3) and (4), which examine the moderating effects of marketing capabilities and technological capabilities on the relationship between policy support and internationalization breadth. The results show that the regression coefficients for Policy×MC and Policy×TC are -3.071 and -13.847, respectively, both statistically significant at the 1% and 5% levels with negative signs. This indicates that marketing capabilities and technological capabilities exert negative moderating effects on the positive relationship between policy support and internationalization breadth. Specifically, the stronger the marketing capabilities and technological capabilities, the weaker the positive association between policy support and internationalization breadth becomes.

To visually illustrate these moderating effects of dynamic capabilities, Figures 2 and 3 plot the moderating roles of marketing capabilities and technological capabilities, respectively, on the policy-internationalization breadth relationship. As depicted in Figure 2, enhanced marketing

capabilities diminish the promotional effect of lower levels of policy support on firms' internationalization breadth, thereby negatively moderating the policy-internationalization breadth relationship. Figure 3 demonstrates that technological capabilities in traditional Chinese medicine enterprises similarly exert a negative moderating effect on the positive relationship between policy support and internationalization breadth. In conclusion, these findings provide support for Hypothesis 3.

[Insert Figure2 about here]

[Insert Figure3 about here]

Models (5) and (6) examine the moderating effects of marketing capabilities and technological capabilities on the relationship between industry competition intensity and internationalization breadth. The results reveal that the regression coefficients for $\text{Indlen} \times \text{MC}$ and $\text{Indlen} \times \text{TC}$ are 31.791 and 158.852, respectively, both statistically significant at the 5% level with positive signs. This indicates that marketing capabilities and technological capabilities exert positive moderating effects on the positive association between industry competition intensity and internationalization breadth. Corresponding moderating effect diagrams for marketing capabilities and technological capabilities on the industry competition intensity-internationalization breadth relationship are presented in Figures 4 and 5. As illustrated in Figure 4, marketing capabilities in traditional Chinese medicine (TCM) enterprises positively moderate the relationship between industry competition intensity and internationalization breadth. Similarly, Figure 5 demonstrates that enhanced technological capabilities in TCM enterprises strengthen the positive relationship between industry competition intensity and internationalization breadth. In conclusion, these findings collectively support Hypothesis 4.

[Insert Figure4 about here]

[Insert Figure5 about here]

4.3 Robustness test

4.3.1. Extended variables

To further ensure the robustness of our findings, this study conducts robustness checks by incorporating additional control variables. Drawing on established methodologies in prior research (Aini et al.,2024; Yan et al.,2024), we introduce two new control variables: CEO overseas background(*CEO ground*) and firm growth (*Grow*).CEO overseas background is measured by whether the CEO has overseas work or educational experience (assigned a value of 1 if yes, 0 otherwise), and growth is measured by the revenue growth rate (the ratio of the difference between this year's revenue and last year's revenue to this year's revenue). The specific regression results are shown in Table 3: After adding the two control variables of CEO overseas background (*CEOground*) and growth (*Grow*), the research results remain consistent with the previous findings, indicating robustness.

[Insert Table3 about here]

4.3.2 Instrumental variables approach

To enhance the empirical reliability, we address endogeneity concerns through an instrumental variables (IV) approach. Valid instruments must satisfy both relevance (strong correlation with endogenous variables) and exclusion restriction (uncorrelated with error terms). Drawing on Aini et al. (2024), we implement a two-stage least squares (2SLS) estimation using the natural logarithm of provincial medical expenditure (*lnME*) as our primary instrument. As reported in Table 4: (1) First-stage results confirm the instrument's statistical significance (coefficient = 0.00567; $p < 0.01$), satisfying the relevance condition; (2) Second-stage estimates maintain a significantly positive relationship ($\beta = 9.315$; $p < 0.05$) between industry competition intensity (*Indlena*) and internationalization breadth, consistent with baseline findings. Moreover, we employ provincial TCM hospital count (*TCM Num.*) as an alternative instrument, grounded in provincial pro-TCM

policy environments. Table 4 confirms: (3) First-stage significance ($p < 0.01$) validates instrument strength; (4) Second-stage results preserve the significantly positive policy effect ($\beta = 0.467$; $p < 0.01$) on internationalization breadth. These robustness checks collectively reinforce our core conclusions' resilience to endogeneity threats and omitted variable bias.

[Insert Table4 about here]

5 Discussion and conclusions

5.1 Key findings

Based on the push-pull theory, this study examines how policy support and industry competitive intensity jointly shape the breadth of internationalization of China's A-share listed traditional Chinese medicine (TCM) firms over the period 2012–2021, with particular attention to the moderating roles of marketing and technological capabilities. The results show that both policy support and industry competitive intensity significantly promote internationalization breadth, albeit through distinct mechanisms. Specifically, policy support alleviates the liability of foreignness by strengthening host-country market pull forces, whereas industry competitive intensity stimulates firm-level international expansion through market selection dynamics.

More importantly, the findings reveal asymmetric moderating effects of firm capabilities. Marketing and technological capabilities negatively moderate the relationship between policy support and internationalization breadth, while positively moderating the relationship between industry competitive

intensity and internationalization breadth. This dual-directional moderation underscores the context-dependent nature of capability deployment and highlights critical boundary conditions under which firm capabilities facilitate—or constrain—international expansion.

5.2 Theoretical implications

Although prior research has highlighted the importance of policy support and industry competitive pressure in shaping the internationalization of emerging-market multinational enterprises (EMNEs), these studies have largely treated these drivers as functionally equivalent or complementary forces, without systematically comparing their underlying mechanisms (Cuervo-Cazurra et al., 2018; Ramamurti & Hillemann, 2018). By contrast, our findings demonstrate that policy support and industry competition intensity exert fundamentally different influences on internationalization breadth, thereby advancing a more nuanced understanding of how external drivers operate in heterogeneous internationalization contexts.

A key theoretical implication of this study lies in its contextualized re-examination of internationalization motives. Owing to the culturally embedded and institutionally complex nature of traditional Chinese medicine (TCM), the internationalization behavior of TCM firms deviates markedly from that of both developed-market MNEs (DMNEs) and conventional EMNEs. Existing theories, which predominantly emphasize strategic asset seeking or institutional arbitrage motives, are therefore insufficient to

explain TCM firms' international expansion. Drawing on push–pull theory and recent calls for greater contextual sensitivity in international business research (Welch et al., 2018; Meyer et al., 2020), this study develops an integrative analytical framework that jointly incorporates home-country push forces and firm-specific capabilities. In doing so, we clarify how home-country pressures shape firms' exposure to intermediate institutional barriers while host-country pull forces generate heterogeneous incentives for international expansion. This perspective extends prior internationalization models by explicitly theorizing the differentiated pathways through which institutional and market forces influence firm behavior.

Moreover, this study contributes to the growing literature on firm capabilities and international strategy by revealing their contingent and asymmetric roles. While marketing and technological capabilities have long been recognized as critical drivers of international competitiveness (Knight & Liesch, 2016; Morgan et al., 2019), our findings show that their effects depend critically on the nature of the external driver. Capability-advantaged TCM firms are able to overcome culturally rooted institutional barriers and transfer ownership advantages across borders, thereby relying less on policy support. In contrast, firms with weaker capabilities depend more heavily on home-country policy support, which serves both as a source of resource augmentation and as a mechanism of legitimacy endorsement in foreign markets (Zhou & Guillén, 2016; Cuervo-Cazurra, 2022). This capability–

institution substitution effect advances EMNE theory by specifying when firm capabilities can replace external institutional support in mitigating the liability of foreignness.

Finally, by demonstrating that marketing and technological capabilities positively condition the impact of industry competitive pressure—but negatively moderate the effect of policy support—this study enriches existing research on competitive dynamics and internationalization. Industry competition operates primarily through market selection mechanisms that reward firms capable of leveraging superior resources and capabilities, whereas policy support functions through institutional mediation. This distinction highlights the importance of aligning capability deployment with the dominant external driver, thereby offering a more dynamic and mechanism-based explanation of strategic choice in international expansion. Overall, our findings respond to recent calls for greater theoretical precision in EMNE research (Luo & Tung, 2018; Verbeke & Kano, 2023) and contribute to a deeper understanding of how firms navigate high-liability-of-foreignness environments through the interaction of institutions, competition, and firm-specific capabilities.

5.3 Practical implications

Firstly, TCM enterprises should recognize that internationalization challenges stem from the joint influence of formal and informal institutional distances between host countries and China , which is manifested as TCM

cultural disparity. Firms must therefore prudently assess risks arising from the combined effects of industry regulatory gaps and national cultural differences during initial international expansion. Building on this institutional understanding, enterprises should strategically prioritize their internationalization pathway selection.

Secondly, TCM enterprises must recognize the distinctiveness of their internationalization challenges , which are fundamentally different from both Chinese firms in conventional sectors and developed-market multinational contexts. Consequently, internationalization templates from these domains prove non-transferable; firms must develop bespoke strategies calibrated to their unique institutional and capability profiles. Crucially, to navigate TCM cultural disparity, we prescribe: that high-capability firms should leverage dual technological and marketing advantages for proactive market expansion, while low-capability enterprises ought to pursue strategic embeddedness in national policies to capture institutional opportunities. This capability-differentiated approach enables firms to secure host-market legitimacy through culturally-attuned positioning, thereby enhancing market penetration in high-disparity regions and amplifying global influence despite institutional complexities—ultimately transforming cultural constraints into competitive differentiation.

Thirdly, at the national level, TCM enterprises' internationalization is substantially bolstered by state policy support. The effective mitigation of

institutional barriers to market access and the offsetting of resource disadvantages are two key benefits of strategic national priorities. It is therefore recommended that policymakers consider increasing the scale of strategic interventions to foster conducive environments, with a particular focus on the needs of SMEs. The enhancement could be achieved through the following measures: The following three points are to be considered: firstly, the enhancement of intergovernmental cooperation in diplomatic engagement, global health governance, and multilateral investment frameworks; secondly, the strengthening of Belt and Road Initiative (BRI) health corridors to amplify Traditional Chinese Medicine's (TCM) global reach; and thirdly, the optimization of the regulatory environment in culturally resistant markets to elevate TCM's international discourse power. Collectively, these measures will underpin sustainable international expansion for TCM multinationals by transforming institutional advocacy into a competitive advantage.

Data Availability statement :

The data supporting the findings of this study are available as supplementary materials associated with this article. The datasets are provided in Stata (.dta) format. Additional materials are available from the corresponding author upon reasonable request.

Ethics statement :

This article does not contain any studies with human participants performed by any of the authors.

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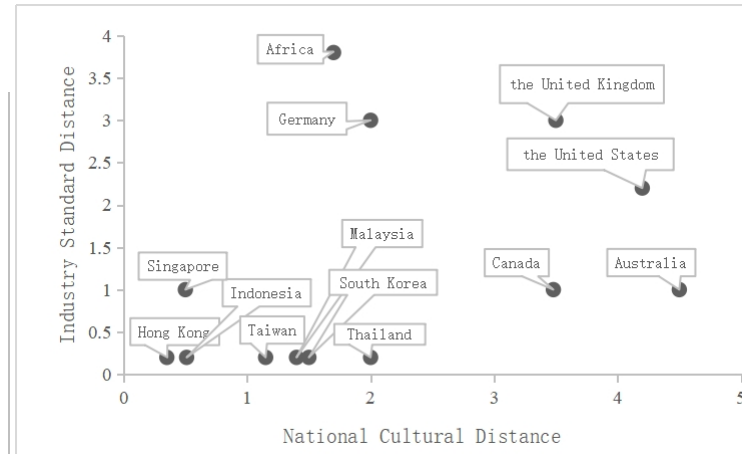


Figure 1 TCM cultural differences faced by TCM multinational enterprises based on the disadvantage of foreignness

Note: Industry norm distance data is calculated based on the differences in the degree of legislation on traditional Chinese medicine between each host country (region) and China, as per Zhang (2022) research. National cultural distance data is calculated based on the six-dimensional scores of Hofstede (2001).

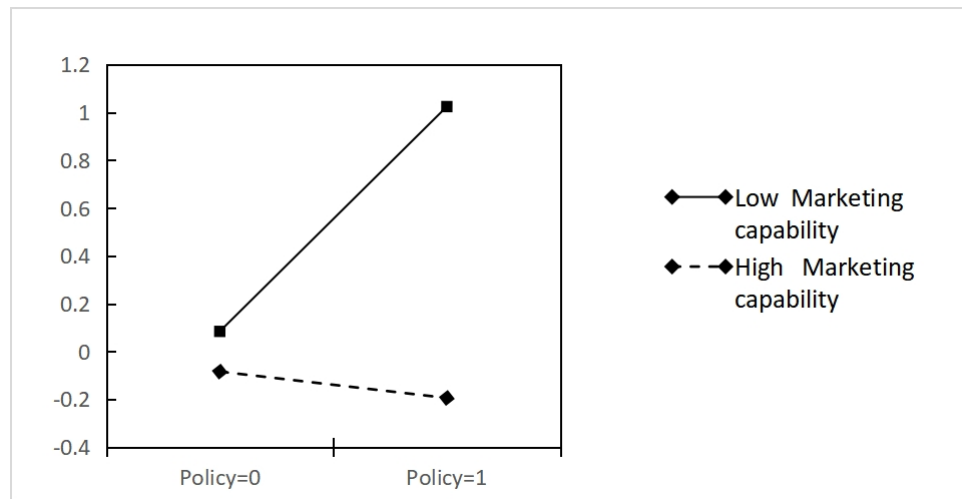


Figure 2 The moderating effect of marketing capability on the relationship between policy support and internationalization breadth

Table 1 Descriptive Statistics and Correlation Analysis																		
Variable	Mean	Standard deviation	Inbre	Policy	Indlen	MC	TC	Age	Size	Lev	ROA	ATO	Tar	Top10	TMT Age	Dual	Board Scale	Indep
Inbre	1.082	2.544	1															
Policy	0.614	0.487	0.102**	1														
Indlen	0.837	0.041	0.122***	-0.506***	1													
MC	0.368	0.171	-0.186***	0.0250	-0.109**	1												
TC	0.036	0.023	-0.0470	0.099**	-0.208***	0.0350	1											
Age	3.002	0.266	0.0730	0.415***	-0.263***	0.0330	-0.077*	1										
Size	21.999	0.992	0.259***	0.128***	-0.192***	0.140***	-0.114**	0.291***	1									
Lev	0.299	0.152	0.133***	-0.0200	0.0610	-0.139***	-0.150***	0.226***	0.298***	1								
ROA	0.067	0.058	-0.0570	-0.199***	0.0690	0.300***	-0.0490	-0.090**	0.153***	-0.288***	1							
ATO	0.689	0.784	0.370***	-0.117***	0.473**	-0.121***	-0.277***	-0.0560	-0.0110	0.0460	0.0610	1						
Tar	0.910	0.085	-0.118***	0.0230	0.0580	0.218***	-0.0670	0.198***	-0.0270	-0.162***	0.162***	0.0340	1					
Top10	56.683	15.483	-0.0710	0.0470	-0.105**	0.186***	0.098**	-0.271***	0.0210	-0.104**	0.226***	-0.120***	0.0520	1				
TMT Age	51.020	7.074	0.0100	0.167***	-0.235***	0.159***	0.139***	0.083*	0.0530	-0.190***	0.128***	-0.209***	0.0600	0.151***	1			
Dual	0.289	0.454	0.077*	0.00500	-0.0350	-0.112**	0.232***	0.0660	-0.229***	0.0180	0.0340	0.0410	-0.0500	0.081*	0.309***	1		
Boardscale	2.169	0.181	0.128***	-0.084*	0.086*	0.0500	-0.263***	0.0200	0.260***	0.210***	0.0120	0.092**	-0.109**	-0.0630	-0.00400	-0.178**	1	
Indep	0.368	0.048	-0.0360	0.0470	-0.176***	-0.0250	0.101**	0.0260	0.0530	0.0640	-0.0330	-0.100**	0.0100	0.125***	-0.0570	0.0310	-0.561***	1

Note: The sample size is 498. ***, **, and * respectively indicate significance at the 1%, 5%, and 10% significance levels.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
<i>Policy</i>		0.467** (0.212)	0.415* (0.212)	0.446** (0.214)			
<i>Indlena</i>		9.315*** (3.568)			8.767** (3.581)	10.277*** (3.614)	
<i>Policy</i> × <i>Indlena</i>							17.02** * (3.577)
<i>MC</i>			-0.492 (0.896)		-0.315 (0.903)		
<i>TC</i>				-2.231 (5.709)		-1.772 (5.743)	
<i>Policy</i> × <i>MC</i>			-3.071** * (0.779)				
<i>Policy</i> × <i>TC</i>				-13.847* * (5.381)			
<i>Indlena</i> × <i>MC</i>					31.791** (13.461)		
<i>Indlena</i> × <i>TC</i>						158.852* * (80.385)	
<i>Age</i>	0.964 (0.764)	1.020 (1.016)	-0.059 (0.903)	.059 (0.944)	1.973** (0.887)	2.377*** (0.898)	0.785 (0.746)
<i>Size</i>	0.51* (0.3)	0.524* (0.302)	0.497* (.297)	0.341 (.302)	0.701** (0.302)	0.598** (0.302)	0.401 (0.293)
<i>Lev</i>	0.713 (0.859)	0.525 (0.858)	0.495 (0.847)	0.842 (0.863)	0.205 (0.862)	0.339 (0.867)	0.801 (0.837)
<i>ROA</i>	0.318 (1.429)	0.726 (1.423)	1.059 (1.413)	0.282 (1.441)	0.516 (1.415)	0.214 (1.425)	0.350 (1.394)
<i>ATO</i>	-2.536* ** (.183)	-2.691** * (.188)	-2.498** * (.181)	-2.53*** * (.186)	-2.499*** * (.201)	-2.611*** * (.195)	-2.413* ** (0.181)
<i>Tar</i>	-4.148* ** (1.453)	-4.048** * (1.451)	-3.715** * (1.444)	-4.508** * (1.447)	-3.566** * (1.448)	-3.83*** * (1.449)	-2.574* * (1.455)
<i>Top10</i>	-0.022* (0.012)	-0.03** (0.012)	-0.025** (0.012)	-0.026** (0.012)	-0.029** (0.012)	-0.029** (0.012)	-0.0239 (0.0117) * (0.0135) * (0.0117)
<i>TMTAge</i>	0.007 (0.014)	0.004 (0.014)	0.006 (0.013)	0.006 (0.014)	0.006 (0.014)	0.005 (0.014)	-0.0019 (0.0135) * (0.0117)
<i>Dual</i>	0.769** * (0.227)	0.809*** * (0.225)	0.763*** * (0.225)	0.786*** * (0.226)	0.787*** * (0.226)	0.761*** * (0.225)	0.740** * (0.222)
<i>BoardScale</i>	1.677* (0.928)	1.871** (0.92)	1.32 (0.915)	1.575* (0.921)	1.708* (0.922)	1.748* (0.924)	1.764* (0.905)
<i>Indep</i>	2.348 (2.749)	3.169 (2.729)	0.089 (2.774)	2.168 (2.744)	1.401 (2.805)	2.549 (2.751)	3.149 (2.686)
<i>Constant</i>	-11.583 * (6.602)	-20.202* * (8.769)	-6.858 * (6.937)	-4.467 * (7)	-25.715** * (8.437)	-26.106** * (8.489)	-9.929 * (6.447)
<i>Individual fixed effects</i>							
<i>Observations</i>	498	498	498	498	498	498	498
<i>R-squared</i>	0.461	0.476	0.487	0.476	0.477	0.475	0.489

Note: ***, **, and * respectively indicate significance at the 1%, 5%, and 10% significance levels. The numbers in parentheses represent standard errors.

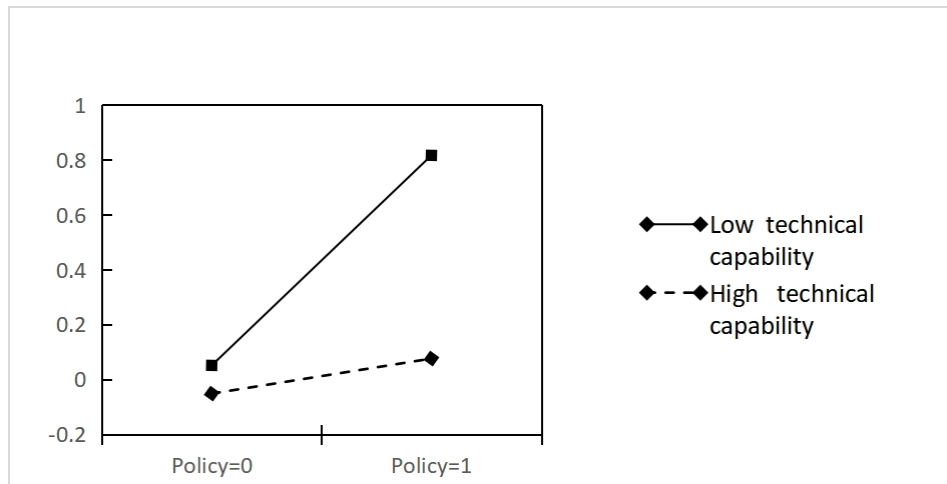


Figure 3 The moderating effect of technical capability on the relationship between policy and internationalization breadth

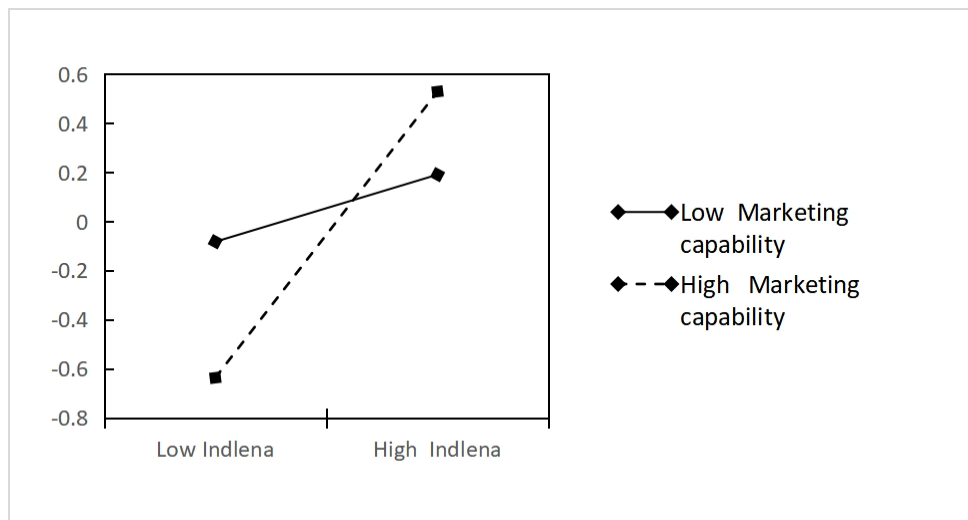


Figure 4 The moderating effect of marketing capability on the relationship between competitive intensity and internationalization breadth

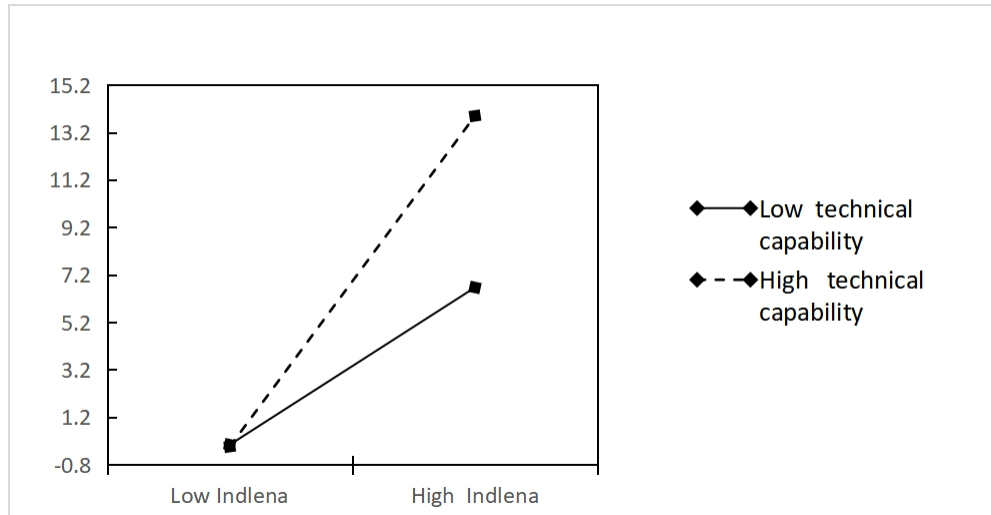


Figure 5 The moderating effect of technical capability on the relationship between competition intensity and internationalization breadth

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	<i>Inbre</i>	<i>Inbre</i>	<i>Inbre</i>	<i>Inbre</i>	<i>Inbre</i>	<i>Inbre</i>	<i>Inbre</i>
<i>Age</i>	1.349* (0.741)	0.469 (0.893)	1.518 (0.994)	0.512 (0.884)	0.609 (0.920)	2.324*** (0.866)	2.592*** (0.876)
<i>Size</i>	0.576** (0.291)	0.509* (0.293)	0.581** (0.293)	0.555* (0.290)	0.442 (0.294)	0.705** (0.292)	0.640** (0.292)
<i>Lev</i>	0.388 (0.834)	0.474 (0.834)	0.264 (0.834)	0.238 (0.828)	0.507 (0.842)	0.033 (0.836)	0.126 (0.842)
<i>ROA</i>	0.314 (1.372)	0.600 (1.378)	0.644 (1.371)	0.9080 (1.367)	0.290 (1.391)	0.490 (1.364)	0.250 (1.371)
<i>ATO</i>	-2.119** *	-2.152** *	-2.266***	-2.132** *	-2.139***	-2.119** *	-2.187***
<i>Tar</i>	(0.19) -3.519** (1.399)	(0.191) -3.76*** (1.402)	(0.196) -3.434** (1.401)	(0.189) -3.252** (1.399)	(0.194) -3.84*** (1.401)	(0.205) -3.074** (1.398)	(0.201) -3.271** (1.397)
<i>Top10</i>	-0.025** (0.012)	-0.028** (0.012)	-0.032*** (0.012)	-0.027** (0.012)	-0.028** (0.012)	-0.030** (0.012)	-0.031*** (0.012)
<i>TMTAge</i>	-0.006 (0.013)	-0.007 (0.013)	-0.008 (0.013)	-0.005 (0.013)	-0.006 (0.013)	-0.007 (0.013)	-0.007 (0.013)
<i>Dual</i>	0.835*** *	0.860** *	0.857***	0.829** *	0.845***	0.841***	0.818***
<i>BoardScale</i>	(0.219) 1.729* (0.911)	(0.219) 1.743* (0.909)	(0.218) 1.951** (0.908)	(0.219) 1.432 (0.905)	(0.219) 1.641* (0.909)	(0.219) 1.870** (0.910)	(0.218) 1.861** (0.911)
<i>Indep</i>	2.862 (2.647)	3.036 (2.642)	3.591 (2.638)	0.933 (2.695)	2.693 (2.656)	2.207 (2.709)	3.119 (2.658)
<i>CEOground</i>	-3.162** *	-3.082** *	-3.045***	-2.856** *	-2.967***	-3.062** *	-3.087***
<i>Grow</i>	(0.530) 0.029 (0.022)	(0.531) 0.029 (0.022)	(0.528) 0.021 (0.022)	(0.531) 0.025 (0.022)	(0.532) 0.028 (0.022)	(0.528) 0.015 (0.022)	(0.527) 0.020 (0.022)
<i>Policy</i>		0.363* (0.207)	0.363* (0.205)	0.332 (0.205)	0.349* (0.208)		
<i>Indlena</i>			8.172** (3.477)			7.937** (3.484)	9.057** (3.52)
<i>MC</i>				-0.657 (.868)		-0.519 (0.871)	
<i>Policy×MC</i>				-2.451** *			
<i>IC</i>				(0.762)	-1.611		-.932

<i>Policy</i> × <i>IC</i>					(5.509) -10.797*		(5.53)
					*		
<i>Indlena</i> × <i>MC</i>					(5.218)	24.401*	
<i>Indlena</i> × <i>I</i> <i>C</i>						(13.202)	142.094*
<i>Constant</i>	-14.316*	-10.139	-22.256**	-10.098	-8.600	-26.603*	(77.427) -27.038**
	*		*			**	*
	(6.354)	(6.770)	(8.480)	(6.737)	(6.792)	(8.159)	(8.198)
<i>N</i>	497	497	497	497	497	497	497
<i>R-squared</i>	0.506	0.51	0.516	0.522	0.515	0.517	0.517
Note: ***, **, and * respectively indicate significance at the 1%, 5%, and 10% significance levels. The numbers in parentheses represent standard errors.							

Table 4 Instrumental Variables Method				
<i>VARIABLES</i>	(1)	(2)	(3)	(4)
	First	Second	First	Second
<i>Indlena</i>	Indlena	Inbre	Policy	Inbre
<i>Policy</i>		58.95** (28.22)		1.923** (2.034)
<i>lnME</i>	0.00567*** (0.00140)			
<i>TCM Num.</i>			0.001*** (5.998)	
<i>Age</i>	-0.132*** (0.0176)	8.398* (4.371)	0.885*** (10.987)	-1.627* (-1.705)
<i>Size</i>	-0.0120*** (0.00434)	1.135 (0.696)	0.038* (1.702)	0.698*** (5.278)
<i>Lev</i>	0.0313** (0.0148)	-1.136 (1.540)	-0.483*** (-3.392)	0.585 (0.611)
<i>ROA</i>	-0.00277 (0.0163)	0.561 (1.417)	-2.008*** (-5.790)	-1.287 (-0.449)
<i>ATO</i>	0.0138*** (0.00162)	-3.338*** (0.553)	0.005 (0.199)	1.267*** (9.281)
<i>Tar</i>	-0.0441*** (0.0163)	-1.676 (3.290)	-0.374 (-1.637)	-2.078 (-1.508)
<i>Top10</i>	0.000489** (0.000187)	-0.0532 (0.0355)	0.006*** (4.914)	-0.015 (-1.610)
<i>TMTAge</i>	0.000157 (0.000198)	-0.000147 (0.0223)	0.011*** (3.656)	0.005 (0.296)
<i>Dual</i>	-0.000119 (0.00319)	0.804** (0.330)	-0.088* (-1.930)	0.815*** (3.049)
<i>ln_BoardScale</i>	-0.0149 (0.0176)	2.748** (1.351)	-0.266** (-1.993)	1.038 (1.290)
<i>Indep</i>	-0.0495 (0.0453)	5.988 (4.331)	-0.570 (-1.162)	1.075 (0.381)

<i>Constant</i>	1.446*** (0.0738)	-99.66** (40.23)	-2.570*** (-4.374)	-11.934*** (-3.105)
<i>Observations</i>	495	498	496	496
<i>R-squared</i>	0.854	0.227	0.348	0.213
Note: ***, **, and * respectively indicate significance at the 1%, 5%, and 10% significance levels. The numbers in parentheses represent standard errors.				

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